

Members	Members	Group #	Title
Andrew	Russ	Group 1	Blue Ammonia
Megan	Carlson	Group 1	
Ameena	Beg	Group 2	Waste Plastic to Kerosene
Giselle	Morales	Group 2	
Asterie	Asterie Sandrine	Group 2	
Jake	Swalec	Group 2	
Ogechikanma	Ihenacho	Group 3	The Chemistry of Clean: Linear AlkylBenzene Production
Shraddha	Zina	Group 3	
Anna	Broviak	Group 3	
Gerrit	Decker	Group 3	
Astrid	Vargas	Group 4	PrintRX Solutions: The Future of Medicine
Emina	Abdic	Group 4	
Andrew	Murillo	Group 4	
Ranim	Bajes	Group 4	
Andrew	Merck	Group 5	Manure to Methanol
Faez	Islam	Group 5	
Ben	Dobbins	Group 5	
Edgar	Castaneda	Group 5	
Danny	Angulo	Group 6	Leftovers to Lift-off: Wood Waste to Jet Fuel
Maxum	Caesar	Group 6	
Cathal	Burke	Group 7	Green Methanol: A New Hope
Matthew	Binder	Group 7	
Angelo	Giles	Group 7	
Jailin	Turner	Group 8	Revolutionizing Pharma: Continuous Production of Acetaminophen
Sofia	Capron	Group 8	
Melina	Savaglio	Group 8	
Shray	Patel	Group 8	
Miriam	Jabr	Group 9	Methane Pyrolysis
Zain	Alsalhani	Group 9	
Gary	Ko	Group 9	
Jacob	Miller	Group 9	
Emily	Seriruk	Group 10	Tires to Fuel & Flavors
Shayma	Abdallah	Group 10	
Jessica	Fajardo	Group 10	
Becky	Listiawan	Group 10	
Karim	Al Zahabi	Group 11	From Grit to Gigabytes: Production of 6N-Purity Silane Gas
Ramiro	Maldonado	Group 11	
Monica	Rochnowski	Group 11	
Lenka	Kostadinov	Group 12	✓ BioDiesel-icious ✓
Mariam	Saeed	Group 12	
Darshil	Patel	Group 12	
Kevin	Malec	Group 12	
Zach	Hinton	Group 13	✓ Green Ammonia ✓
Zuhai	Wang	Group 13	
Gabe	Nkpe	Group 13	
Billy	Kropke	Group 13	
Aman	Dudeja	Group 14	✓ Sweetening the Deal: Glucose's Transformation into Adipic Acid ✓
Emily	Azimi	Group 14	
Shaheer	Sohail	Group 14	
Emmanuel	Risasi	Group 14	
Nit	Patel	Group 14	

UIC ChE Design Showcase Judging Schedule and Assignments

First Name	Last Name	Former/Current Employer	1:00 PM	1:30 PM	2:00 PM	2:30 PM
Alena	Morabito	Baxter	Team #1	Team #2	Team #3	Team #4
Patricia	Gillenwater	AIChE/Nalco	Team #4	Team #1	Team #2	Team #3
Adam	Kanyuh	Honeywell UOP	Team #3	Team #4	Team #1	Team #2
Julieta	Rodriguez	Tetra Pak	Team #2	Team #3	Team #4	Team #1
Paul	Diaz	Constellation	Team #5	Team #6	Team #7	Team #8
Sajjid	Odeh	BakerRisk	Team #8	Team #5	Team #6	Team #7
Robert	Orent	Stepan Company	Team #7	Team #8	Team #5	Team #6
Sedra	Khan	Abbvie	Team #6	Team #7	Team #8	Team #5
Basit	Owoseni	Mondoléz	Team #9	Team #10	Team #11	Team #12
Jesse	Anderson	UL Solutions	Team #13	Team #9	Team #10	Team #11
Joseph	Tijerina	ComEd	Team #11	Team #12	Team #9	Team #10
Renata	Swarbrick	BakerRisk	Team #10	Team #11	Team #12	Team #9
Danielle	Flagg	Sargent & Lundy	Team #13	Team #14	Team #1	Team #5
Tim	Politano	LanzaTech	Team #2	Team #13	Team #14	Team #6
Jeffrey	Lullo	TM Process & Controls, Inc.	Team #7	Team #3	Team #13	Team #14
Stefanie	Broviak	Baxter	Team #14	Team #4	Team #8	Team #13
Caleb	Tristano	Constellation	Team #9	Team #10	Team #11	Team #12
Arwa	Hasan	Medline	Team #12	Team #13	Team #14	Team #11
Dave	Modeste	LanzaJet	Team #6	Team #8	Team #5	Team #7
Jeffrey	Zalc	BP	Team #10	Team #14	Team #12	Team #9
Rija	Firoz	Constellation	Team #3	Team #2	Team #4	Team #1

Handwritten initials

2024 CHEME EXPO JUDGING SHEET

Judge Name: Paul Diaz

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	3
Total	21

Comments:

Loved the pink Theme for the group! Project scope was well defined and ~~for~~ focused, Process was easy to follow despite chemicals being very specialized. More room ^{in project scope} to discuss byproducts, would've been interesting.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Paul Diaz

Team Number: 5

Team Members:

3

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	17

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	3
Total	18

Comments:

Good Group dynamic. Asked for feedback and thoughts on Safety analysis.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Paul Diaz

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	2
Total	13

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	

Comments:

For a group of 2 people, the process was too ambitious. Good focus on tax credits to potentially open the market favorably.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Paul Diaz

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	2
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	12

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	3
3. Organization & quality of presentation material	2
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	2
Total	13

Comments:

Process was confusing despite small scale of facility.
Choosing China for facility/economic market was
a confusing choice.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Arwa Hasan

Team Number: 14

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3 → project objectives are not very clear, what
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	2
5. Quality, effectiveness, & use of visual aids	4
Total	16

Comments:

Technical Quality: project objective wasn't clear, what is adipic acid used in? Customer base/Markets. Location is not defined in Texas for water based process.

Presentation: one person was doing 90% of the presenting and others tried to participate.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Arwa Hasan

Team Number: 11

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

- good presentation
- project wasn't very sustainable, not much consideration to CO₂ emission, if CO₂ emission was out of scope in the project, that was not clear

2024 CHEME EXPO JUDGING SHEET

Judge Name: Arwa Hasan

Team Number: ~~18~~ 12

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4 - overall good knowledge
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Arwa Hasan

Team Number: 12 13

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4 - Cost such as water treatment not evaluated
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4 3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Presentation: was great overall, professional, everyone presented their materials

2024 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni

Team Number: 12

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3 4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3 4
2. Knowledge of the background material	3 4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni

Team Number: 11

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni

Team Number: 10

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	21

Comments:

\$3.1 m
0.7 → 8,100 h₁

2024 CHEME EXPO JUDGING SHEET

Judge Name: Basit Owoseni

Team Number: 9

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3.0 3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4.0
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3.0 3
4. Overall fulfillment of project objectives	3.0 4
5. Project practicality in view of contemporary engineering standards/practice	4.0
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	21

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	28

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	22

Comments:

Water ~ Cat poison / Yucca

Process Control

Large Market

Overall excellent
Presentation's
Team

Carbon

Elac 55 Kwh

Sustain
Small plan
Carbon Balance
Recycle Carbon

2024 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	21

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 5

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	21

Comments:

Disinterested/
Cap cost - Multi
Higher Price to get More Revenue
? Raw Mat'l Cost / Plant Location ✓
HEN.
? Production Target.
? Waste, Solid Lys Gas (CO Capture)
Safety Concerns.
Scale.
? instrumentation?
Process Sump
Condenser
Sump
additional CO₂ Product
Politics Gov

2024 CHEME EXPO JUDGING SHEET

Judge Name: Dave Modeste

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	21

Comments:

- Market 61. growth
- 1. Discount
- HAZOP Crystallizer
- Motor speed Control
- Instrument Analytical
- T pH Xcel
- Recycle a Cost
- Low utility usage

- Save on Cooling Water
- Discount factor
- Charcoal / Dust Consider
- pH
- Materials of Constr

2024 CHEME EXPO JUDGING SHEET

Judge Name: Robert Orent

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	5 3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	16

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	4 3
5. Quality, effectiveness, & use of visual aids	4
Total	17

33

Comments:

- NO scope for COST
- A1+ Elect safety (solar) - ?
- no consideration of project safety (H₂)
 - did perform H₂ exp.
- no flame

2024 CHEME EXPO JUDGING SHEET

Judge Name: Robert Orent

Team Number: 5

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3 21
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	18

Comments:

- understood unit cost - product.
- ~~Good~~ sensitivity
- Good
- decent comments / speaking

(37)

2024 CHEME EXPO JUDGING SHEET

Judge Name: Robert Orent

Team Number: 8

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	3
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	21

market understanding

(41)

Comments:

- Good speakers
- continuous etc is better
↳ novel
- limited waste scope

2024 CHEME EXPO JUDGING SHEET

Judge Name: Robert Orent

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

49

Comments:

- Good Safety
- Good sustainability
 - structure / power generation, integrated
- Good Technical understanding
 - considered flare
 - Good presentation
 - understand Environmental impacts
 - SCR

- understand economic and possible of financial returns

45/50

2024 CHEME EXPO JUDGING SHEET**Judge Name:** Sedra Khan**Team Number:** 8 *Revolutionizing Pharma***Team Members:** *Jailin, Sofia, Medina, Gray***Technical Quality**

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments:

Great Work! Continue to research the market, would want to better understand why the process is better than competitors and overall fantastic job!

2024 CHEME EXPO JUDGING SHEET

40/50

Judge Name: Sedra Khan

Team Number: 7 Green Methanol

Team Members: Matthew, Cathal, Angelo

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids → minor typos in Recommendations section	4
Total	22

Comments:

Great work! would want to hear more about initiatives to improve the costs and process. Understanding the applicability, and purpose is also an area to explore — would have appreciated hearing more about the market analysis completed.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Sedra Khan

Team Number: 6 *leftovers to lift-off*

Team Members: Danny and Maxum

49/50

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

- steam generation → great job answering questions!
- cleaning streams to increase sustainability was a great alternative offered!

Fantastic job presenting — great dynamic and composed responses!

2024 CHEME EXPO JUDGING SHEET

46/50

Judge Name: Sedra Khan

Team Number: 5 *Measure to Methanol*

Team Members: Andrew, Faez, Ben, Edgar

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5 3
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25 23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	23

Comments:

Great Work! Displayed alot of proactive solutions - whenever there was a gap or setback they addressed how it could be resolved! including government incentives would be benefical as well.

Team 4 = Print RX Solutions = The Future of Medicine

2024 CHEME EXPO JUDGING SHEET

Judge Name: Julieta Rodriguez

Team Number: 4

Team Members: Astrid V, Emina A, Andrew M, Ramim B.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	5
Total	22

Comments:

Presentation was well delivered. The model helped a lot to explain their process. Technical difficulties explained very well and technical questions answered with knowledge of the topic.

Team 3 = The Chemistry of Clean- Linear Alkylbenzene Prod

2024 CHEME EXPO JUDGING SHEET

Judge Name: Julieta Rodriguez

Team Number: 3

Team Members: Ogechikanma I, Shraddha Z, Anna B, Genit D.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	18

Comments: The team was well prepared while presenting, and I would've liked to see the complete team interacting more. When challenged about safety concerns, the team was well prepared mentioning the considerations and the way of mitigating those risks.

Team 2: Waste Plastic to Kerosene

2024 CHEME EXPO JUDGING SHEET

Judge Name: Julieta Rodriguez

Team Number: 8

Team Members: Ameenah, Giselle M, Astene S, Jake S.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	3
Total	17

Comments:

The team explained thoroughly the technical & economical aspects of their designs. I would have liked to see more interaction of all the team members.

Team 1: Blue Ammonia

2024 CHEME EXPO JUDGING SHEET

Judge Name: Julieta Rodriguez

Team Number: 0

Team Members: Andrew R, Megan C.

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	14

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	3
Total	20

Comments:

Technically, both students did great by explaining their projects with considerations and challenges. The set back is the location of the plant; not taking into account economic environment is critical. The team could have a stronger argument with green solutions.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Tim Politano

Team Number: 14

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	14

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	2
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	2
5. Quality, effectiveness, & use of visual aids	3
Total	13

Comments: It's very important to know your input/output costs + specifications. ~~Always~~ Start from the outside + work inward, because these will define much of what you do in the process. Check your effect of product cost on NPV; I suspect it has more impact, but could be wrong.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Tim Politano

Team Number: 13

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4/3
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	21

Comments: Well-presented, all. Poster presentations are challenging, especially w/ so much info. density. Consider highlighting Good use of handout for better scale/visibility of fine details, but also consider whether these same elements are readable on the poster or can be modified.

The Alchemy of Air is the book I mentioned. Highly recommend now that you have some contact on annars production

2024 CHEME EXPO JUDGING SHEET

Judge Name: Tim Politano

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	23

Comments: Well-presented by all. Poster presentations are difficult, especially w/ so much info. density. Highlighted key areas well. Good understanding of your process, where things go, how things could change as assumptions meet real-world conditions

2024 CHEME EXPO JUDGING SHEET

Judge Name: Tim Politano

Team Number: 6

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Excellent presentation, very well done! You covered a lot of ground but did so clearly, technically, with solid understanding. Loved the color coding on the PFD to organize the view. Waste heat integration ideas (turbine to elec.; heating boiler feed water from exotherm) were great + highlighted your level of understanding. Appreciated the clear design decision explanation for your gasification tech. Best wishes to you in your future work!

2024 CHEME EXPO JUDGING SHEET

Judge Name: Renata Swarbrick

Team Number: 10

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	24

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Great team dynamics and presentation skills. I appreciated the female leadership in this group.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Renata Swarbrick

Team Number: 11

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Great presentation; solid knowledge of process and market. Good team cohesion.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Renata Swarbrick

Team Number: 12

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Excellent presentation! I was delighted to see female leadership in this group. Great cohesion amongst all team members. Good effort justifying chosen process. Above and beyond visual/presentation aids.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Renata Swarbrick

Team Number: 9

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Great presentation! I appreciated the thorough responses to questions and overall team dynamics.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Joseph Tijerina

Team Number: 9 Methane Pyrolysis

Team Members:

Technical Quality 43

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	21

Comments:

Good job, but lacked some knowledge on general operational points. Presentation was sporadic and hard to follow. Creative!, but hard on the eyes. Impressive presentation, could have used more prep.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Joseph Tijerina

Team Number: 10

Team Members:

Tires to fuel & flavors

Technical Quality *870 49*

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	<i>5</i>
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	<i>5</i>
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	<i>5</i>
4. Overall fulfillment of project objectives	<i>5</i>
5. Project practicality in view of contemporary engineering standards/practice	<i>5</i>
Total	<i>25</i>

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	<i>5</i>
2. Knowledge of the background material	<i>3</i>
3. Organization & quality of presentation material	<i>5</i>
4. Group dynamics/member participation	<i>5</i>
5. Quality, effectiveness, & use of visual aids	<i>4</i>
Total	<i>22</i>

Comments:

Great board, they had answers to every question I had. Well organized, you could tell they went deep on research. Did not have total understanding of how their product could be used in the bio diesel space and didn't solve for their steel bio product issue

Print Rx Solutions: Future of Medicine ^{#4}

2024 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 4

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	18

- great 3D model

Comments:

- Very interesting project very non-traditional versus the other projects
- Liked the 3D model of plant and overall good presentation
- The team had 2 very strong members who seemed to have done all of the work.

Blue Ammonia

2024 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 1

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

Comments:

- Megan & Andrew worked very well together.
- They both had a good understanding of the overall project as well as the economics.
- They provided good insight on what they would do to improve the project economics to make it more profitable.
- Overall the best presentation I saw.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 2

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	4
Total	15

→ did not understand the benefits of renewable jet vs kerosene

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	3
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	16

Comments:

- Team didn't seem to have good chemistry and overall didn't seem to understand their market. ~~why~~ Their pricing and economics were not fully thought out.

Giselle & Ameenah were main contributors and did most of the presenting.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Adam Kanyuh

Team Number: 3

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	5
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	4
Total	17

Comments:

Overall good ~~the~~ knowledge of the process. It was clear their mentor gave them a lot of insight. Their project was not as developed from the entire complex like the other projects I saw.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Joseph Tijerina

Team Number: 11 Grit to Gigabytes: Production of LN - Purity of Silane Gas

Team Members:

Technical Quality 43

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4 5
Total	22

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	3
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	21

Comments: They understood the utilities cost & purity were their limiting factors, did not have a feasible solution to impurity issue, but understood the issue @ hand. Needed a bit of work on presentation transitions speaking up; bit small, they had total knowledge and ability to answer questions. Uninformed about the vs market & how Chinese manufacturing is currently done.

avg

①

12

②

6

③

9

2 ser

6

4

12

2024 CHEME EXPO JUDGING SHEET

Judge Name: Danielle Flagg

Team Number: 13

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	16

Did not look at many options
Did not consider much of the balance of plant.

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	22

a few showed up late.

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Joseph Tijerina *Bridges - 12/005*

Team Number: 12

Team Members:

Technical Quality *4/5 4/1*

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3 <i>4/1</i>
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5 <i>3</i>
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	<i>21</i>

*did not present alternatives with real knowledge
→ hazy about different purity of product & the effects of equipment and yield
↳ Did not have much in place for safety at all. No*

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5 <i>3</i>
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	4/1
5. Quality, effectiveness, & use of visual aids	5 <i>3</i>
Total	<i>20</i>

Comments:

*did not have a plant mock-up.
not every group member participated the same.
They didn't factor in variability of raw material
↳ the plant mock up was incorrect, not the same as in the brochure*

2024 CHEME EXPO JUDGING SHEET

Judge Name: Danielle Flagg

Team Number: 14

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	4
Total	16

Discussed one option considered

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	4
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	2
5. Quality, effectiveness, & use of visual aids	3
Total	15

planned presentation was a little chaotic

only presented for a few minutes and most was questioning.

Emily dominated the presentation but the others did a good job of jumping in.

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Danielle Flagg

Team Number: 1

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	25

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Very well communicated, well researched project, and good collaboration, especially considering only two people.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Danielle Flagg

Team Number: 5

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	18 23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	20 25

Comments:

very well communicated and
good team collaboration

2024 CHEME EXPO JUDGING SHEET**Judge Name:** Jesse Anderson**Team Number:** 10**Team Members:****Technical Quality**

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	21

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	3
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20

Leader, too quiet

Some quiet members

Comments:

Great novel project.

~~But~~ Tire key chain is pretty cool.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Jesse Anderson

Team Number: 11

Team Members:

Production of b-N
Purify silane

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

Extra cost for clean room step?

Waiting for a while.

Entirely dependent on

Blackbox CO₂ capture.

Semiconductors.

Pilot plant, recte distillation [3] innovation, based off papers

Aspen Sim.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Jesse Anderson

Team Number: 9

Team Members:

Two voice - Hydrogen!
A Stepping-stone into the
Future

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	4
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	25

Comments:

Very knowledgeable.
Conservative numbers
High contingency
Very passionate about project!
Brochure printed weird.

2024 CHEME EXPO JUDGING SHEET

Back to Basics: Green Ammonia

Judge Name: Jesse Anderson

Team Number: 1213

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	2
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	5
Total	15

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	2
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	2
Total	14

Comments:

energy for electrolyzer? price of electrolysis & air separation how?
water & O₂ removal how?
variability of Texas
power grid?
small graphics

2024 CHEME EXPO JUDGING SHEET

Judge Name: Caleb Tristano

Team Number: 12

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	5
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	24

Comments:

- good job using Feedstock constraints to drive location (H_2)
- good job recognizing NO_2 gas hazard (Bran Gas?)
- 320°C Hydro cracking
- catalysts: nickel / tungsten.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Caleb Tristano

Team Number: 11

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	4
5. Quality, effectiveness, & use of visual aids	4
Total	20

Comments:

L2 Not a lot of detail about down/outage time to maintain the plant.
- Safety for this process (w/ solids and silica) could use more time and/or poster realstate.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Caleb Tristano

Team Number: 10

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	22

→ missing some stuff
↓
financial/chemistry

Comments:

475C
I don't see the chemistry as side products.

71% TCFEOR.

- Strength & $\vec{B}$
- Sharp metal Howard.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Caleb Tristano

Team Number: 9

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	4
Total	20

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	22

Comments:

relatively low pressure. (+)
Safety for handling of solid carbon?
process in the Rx is highly sensitive to O_2/H_2O , but there's an H_2O HX.

2024 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Lullo

Team Number: 14

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

2	1. Identification of project objective(s)	2
1	2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
1 2	3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	1
2	4. Overall fulfillment of project objectives	2
1 2	5. Project practicality in view of contemporary engineering standards/practice	1
Total		8

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

3	1. General professionalism and oral communication	3
2	2. Knowledge of the background material	2
3	3. Organization & quality of presentation material	3
2	4. Group dynamics/member participation	2
4	5. Quality, effectiveness, & use of visual aids	4
Total		14

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Lullo

Team Number: 13

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

4	1. Identification of project objective(s)	4
3	2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
2	3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
3	4. Overall fulfillment of project objectives	3
2	5. Project practicality in view of contemporary engineering standards/practice	2
Total		14

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

4	1. General professionalism and oral communication	4
3	2. Knowledge of the background material	3
4	3. Organization & quality of presentation material	4
4	4. Group dynamics/member participation	4
4	5. Quality, effectiveness, & use of visual aids	4
Total		19

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Lullo

Team Number: 3

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

4	1. Identification of project objective(s)	4
4	2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3	3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4	4. Overall fulfillment of project objectives	4
4	5. Project practicality in view of contemporary engineering standards/practice	4
Total		19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

4	1. General professionalism and oral communication	4
3	2. Knowledge of the background material	3
3	3. Organization & quality of presentation material	3
4	4. Group dynamics/member participation	4
4	5. Quality, effectiveness, & use of visual aids	4
Total		18

Comments:

- Flare
- 250 acres

2024 CHEME EXPO JUDGING SHEET

Judge Name: Jeffrey Lullo

Team Number: 7

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

4	1. Identification of project objective(s)	4
2	2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2
2	3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2
4	4. Overall fulfillment of project objectives	4
3	5. Project practicality in view of contemporary engineering standards/practice	3
Total		15

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

5	1. General professionalism and oral communication	5
4	2. Knowledge of the background material	4
4	3. Organization & quality of presentation material	4
5	4. Group dynamics/member participation	5
4	5. Quality, effectiveness, & use of visual aids	4
Total		22

Comments:

→ safety
→ cost
→ plant layout

2024 CHEME EXPO JUDGING SHEET

Judge Name: Patricia Gillenwater

Team Number: 4 *Print RX Solutions: Future of Medicine*

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	4
5. Project practicality in view of contemporary engineering standards/practice	3
Total	19

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	5
2. Knowledge of the background material	5
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	3
5. Quality, effectiveness, & use of visual aids	5
Total	22

Comments: *Simulation converged.*

2024 CHEME EXPO JUDGING SHEET

Judge Name: Patricia Gillenwater

Team Number: 3

Team Members: *The Chemistry of Clean: Linear Alkyl Benzene Production*

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	4
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	18

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	4
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	4
Total	21

Comments:

2024 CHEME EXPO JUDGING SHEET

Judge Name: Patricia Gillenwater

Team Number: 2 *Waste Plastic to Kerosene*

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	16

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	2
5. Quality, effectiveness, & use of visual aids	3
Total	15

Comments: *simulation converged
one person barely talked.*

2024 CHEME EXPO JUDGING SHEET

Judge Name: Patricia Gillenwater

Team Number: 1 *Blue Ammonia*

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	3
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	3
4. Overall fulfillment of project objectives	3
5. Project practicality in view of contemporary engineering standards/practice	3
Total	16

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	3
3. Organization & quality of presentation material	3
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	3
Total	18

Comments: *Simulation ~~soft~~ convergence issues.*

CHE Senior Design Showcase Judges' Agenda

Location:

- **Lunch & Instructions:** EIB Room 224
- **Poster Displays:** EIB First Floor Hallway and Room 124
- **End of Year Banquet:** Student Center East, Room 301

Schedule:

- **12:00 PM - 1:00 PM:** Arrival and Lunch
 - Judges arrive and proceed to Room 224
 - Lunch served
 - Distribution of judging sheets and instructions
- **1:00 PM - 3:00 PM:** Judging Segment
 - **1:00 PM - 1:30 PM:** Project Evaluation 1
 - **1:30 PM - 2:00 PM:** Project Evaluation 2
 - **2:00 PM - 2:30 PM:** Project Evaluation 3
 - **2:30 PM - 3:00 PM:** Project Evaluation 4
- **3:00 PM - 5:00 PM:** Break
 - Time for judges to deliberate and finalize scores
- **5:00 PM - 7:00 PM:** CHE End of Year Banquet (Please join us if you are available)
 - Winners announced
 - Networking and celebration

Awards:

- **1st Place:** \$500 sponsored by LanzaJet
- **2nd Place:** \$400 sponsored by Honeywell UOP
- **3rd Place:** \$300 sponsored by Stepan

Special Thanks:

We extend our deepest gratitude to our sponsors, LanzaJet, Honeywell UOP, and Stepan, for their generous support of the CHE Senior Design Showcase.